

3. Chlorine, bromine and iodine are elements in Group 7 of the Periodic Table.

- (a) State the number of electrons in the outer shell of a chlorine atom. [1]

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- (b) Underline the correct formula of chlorine gas. [1]



- (c) A teacher demonstrated the reactivity of Group 7 elements by placing heated iron wool in gas jars containing vapours of the elements.

The class recorded the results shown in the table.

Group 7 element	Observation with heated iron wool
chlorine	glowed very brightly
bromine	
iodine	glowed faintly

- (i) Put a tick (✓) in the box to show the observation when heated iron wool was placed in bromine vapour. [1]

glowed less brightly than iodine

☐

glowed less brightly than chlorine

☐

glowed more brightly than chlorine

☐


- (ii) The compound produced in the reaction between iron wool and bromine contains the ions Fe^{3+} and Br^- .

I. Give the formula of this compound. [1]

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II. State the name of this compound. [1]

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- (d) Tick (✓) the box next to a common use of iodine. [1]

to disinfect skin before surgery

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to make coloured fireworks

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to sterilise swimming pools

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to fill party balloons

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